作业

D8.2 For the circuit of Fig. 8.4, let $V_{DD} = V_{SS} = 1.5$ V, $V_{tn} = 0.6$ V, $V_{tp} = -0.6$ V, all channel lengths = 1 µm, $k_n' = 200\ \mu A/V^2$, $k_p' = 80\ \mu A/V^2$, and $\lambda = 0$. For $I_{REF} = 10\ \mu A$, find the widths of all transistors to obtain $I_2 = 60\ \mu A$, $I_3 = 20\ \mu A$, and $I_5 = 80\ \mu A$. It is further required that the voltage at the drain of Q_2 be allowed to go down to within 0.2 V of the negative supply and that the voltage at the drain of Q_5 be allowed to go up to within 0.2 V of the positive supply.

Ans. $W_1 = 2.5\ \mu m$; $W_2 = 15\ \mu m$; $W_3 = 5\ \mu m$; $W_4 = 12.5\ \mu m$; $W_5 = 50\ \mu m$

对于 Q1: $10\mu A = \frac{1}{2} 200\ \mu A/V^2 \cdot \frac{W_1}{1\mu m} \cdot (0.2V)^2$

$\Rightarrow W_1 = 2.5\mu m$

$\therefore \frac{W_2}{W_1} = \frac{60}{10} \Rightarrow W_2 = 15\mu m$

$\frac{W_3}{W_1} = \frac{20}{10} \Rightarrow W_3 = 5\mu m$

对于 Q4:

$20\mu A = \frac{1}{2} \cdot 80 \cdot \frac{W_4}{1} \cdot (0.2V)^2$

$\Rightarrow W_4 = 12.5\mu m$

$\frac{W_5}{W_4} = \frac{80}{20} \Rightarrow W_5 = 50\mu m$

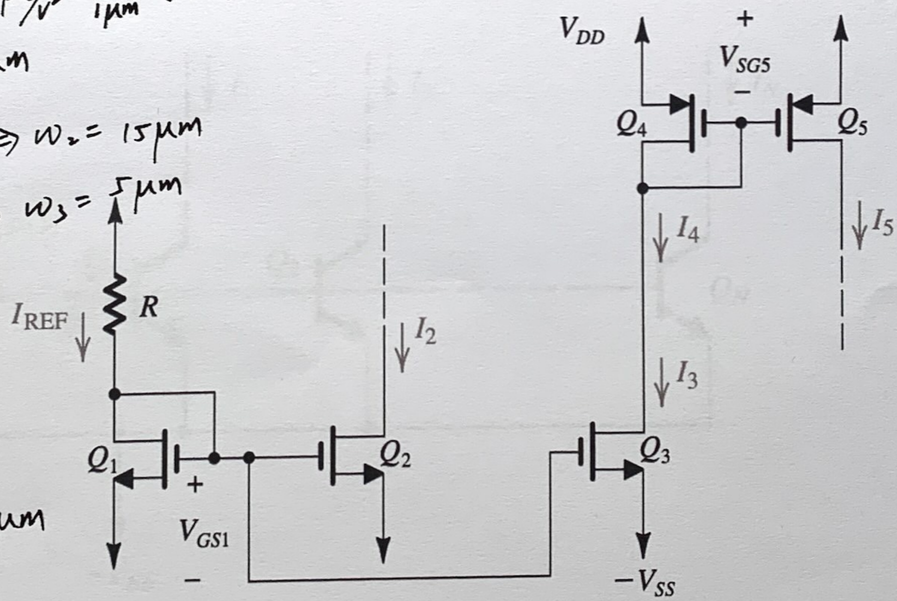


Figure 8.4 A current-steering circuit.

作业

8.5 Figure E8.5 shows an N-output current mirror. Assuming that all transistors are matched and have finite β and ignoring the effect of finite output resistances, show that

$$I_1 = I_2 = \cdots = I_N = \frac{I_{REF}}{1 + (N+1)/\beta}$$

~~For $\beta = 100$, find the maximum number of outputs for an error not exceeding 10%.~~

$$\underset{①}{\frac{I_i}{\beta}} \cdot N + \underset{②}{\frac{I_i}{\beta}} + \underset{③}{I_i} = I_{REF}$$

$$I_i\left(\frac{N}{\beta} + \frac{1}{\beta} + 1\right) = I_{REF}$$

$$I_i = \frac{I_{REF}}{1 + (N+1)/\beta}$$

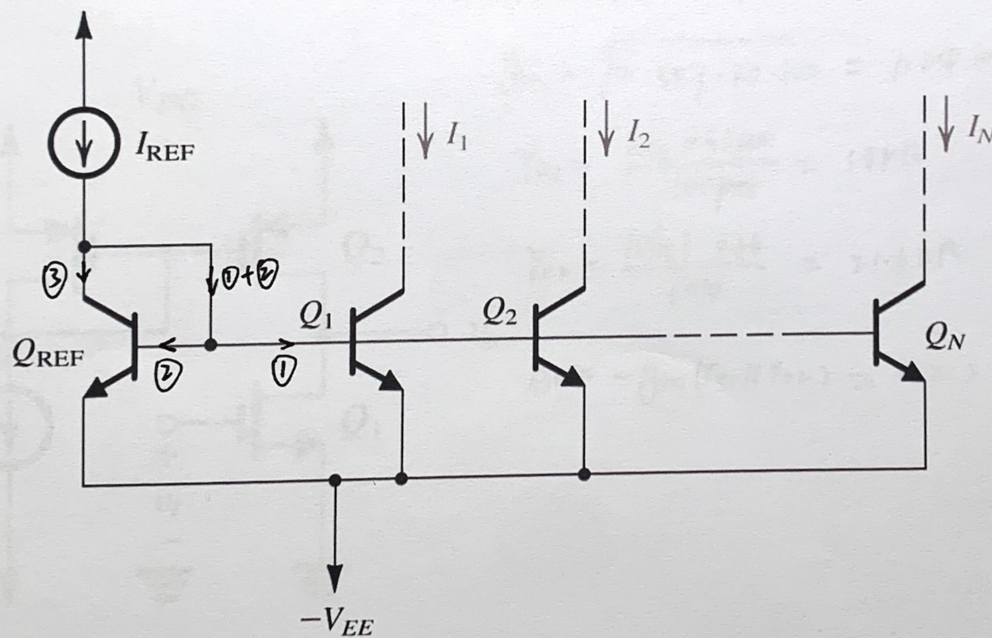
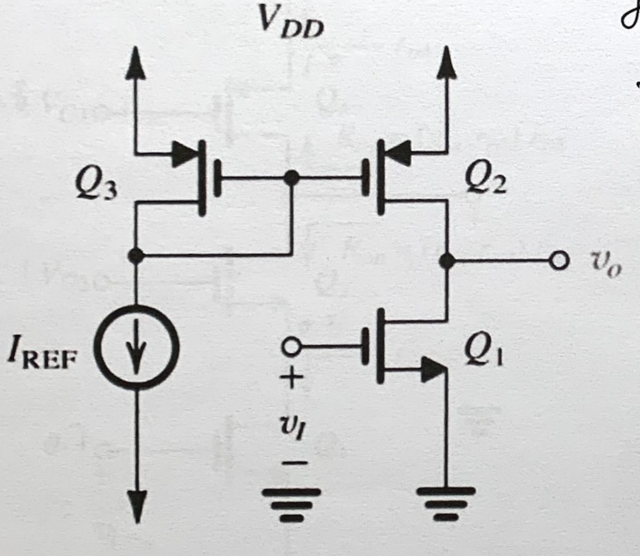


Figure E8.5

~~Ans. 9~~

[redacted]

8.8 A CMOS common-source amplifier such as that in Fig. 8.16(a), fabricated in a 0.18-μm technology, has $W/L = 7.2\ \mu m/0.36\ \mu m$ for all transistors, $k_n' = 387\ \mu A/V^2$, $k_p' = 86\ \mu A/V^2$, $I_{REF} = 100\ \mu A$, $V_{An}' = 5\ V/\mu m$, and $|V_{Ap}'| = 6\ V/\mu m$. Find g_{m1}, r_{o1}, r_{o2}, and the voltage gain.
Ans. 1.24 mA/V; 18 kΩ; 21.6 kΩ; –12.2 V/V



(a)

$I_{D1} = I_{REF} = 100\ \mu A$

$g_{m1} = \sqrt{2 \cdot 387 \cdot 20 \cdot 100} = 1.24\ mA/V$

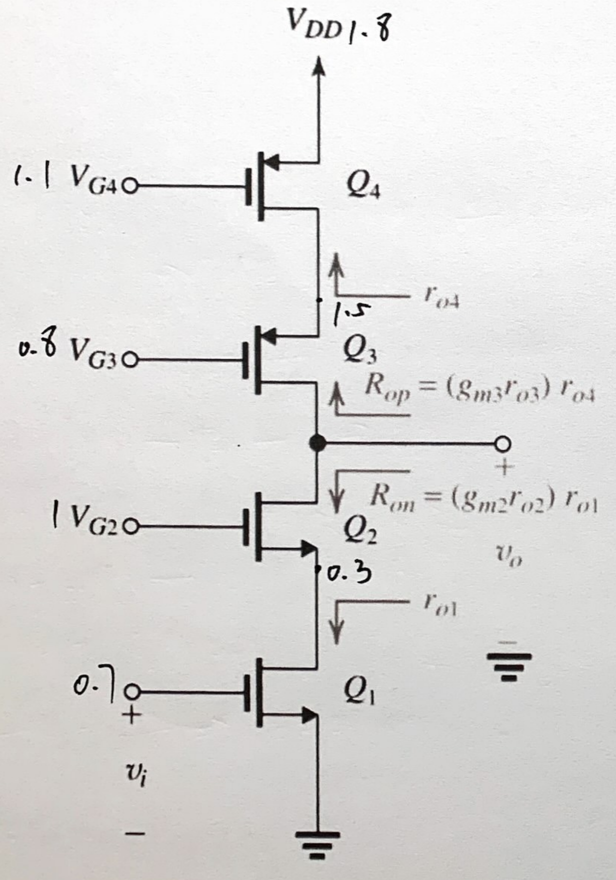
$r_{o1} = \dfrac{V_{An}' \cdot 0.36\ \mu m}{100\ \mu A} = 18\ k\Omega$

$r_{o2} = \dfrac{|V_{Ap}'| \cdot 0.36}{100} = 21.6\ k\Omega$

$A_v = -g_m (r_{o1} \| r_{o2}) = -12.2\ V/V$

8.20 Consider the cascode amplifier of Fig. 8.31 with the dc component at the input, $V_I = 0.7$ V, $V_{G2} = 1.0$ V, $V_{G3} = 0.8$ V, $V_{G4} = 1.1$ V, and $V_{DD} = 1.8$ V. If all devices are matched (i.e., if $k_{n1} = k_{n2} = k_{p3} = k_{p4}$), and have equal $|V_t|$ of 0.5 V, what is the overdrive voltage at which the four transistors are operating? What is the allowable voltage range at the output?

Ans. 0.2 V; 0.5 V to 1.3 V



(a)

$V_{ov} = 0.7 - 0.5 = 0.2V$

根据图，V_{out} 的上限是 $1.5 - 0.2 = 1.3$

下限 $0.3 + 0.2 = 0.5$